EPISODE 2 RECAP

TYPES OF DATA

- BINARY
- CATEGORICAL → NOMINAL: region, class
- CATEGORICAL → ORDINAL: military ranks
- NUMERICAL → COUNT DATA: children, phones
- NUMERICAL → CONTINUOUS DATA: return, weight

VISUALISATION

- few values → • frequency table • bar plot
- many values → • histograms • box plot

LOCATION MEASURES (numerical data)

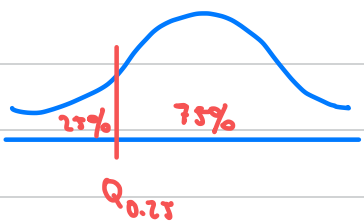
MEAN: $\frac{1}{n}\sum_{i=1}^{n} x_i$

MEDIAN: value in the middle of ordered data (robust wrt extreme values)

QUANTILE OF ORDER e.g. 0.25 (also called 25th percentile, 1st quartile)

↳ ≈ 25% obs fall on its left

≈ 75% obs fall on its right



e.g. 90th percentile of italian salaries (gross, per year) is: 40k

VARIATION MEASURES

RANGE: max − min (or [min, max] as an interval)

INTERQUARTILE RANGE: $Q_3 - Q_1$ (Third Quartile − First Quartile)

VARIANCE: $\frac{1}{n}\sum_{i=1}^{n}(x_i - \text{MEAN})^2$

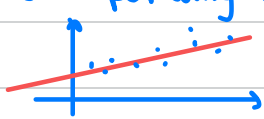
STANDARD DEVIATION: $\sqrt{\text{VARIANCE}}$

MEAN AVERAGE DISTANCE: $\frac{1}{n}\sum_{i=1}^{n}|x_i - \text{MEAN}|$

EXPLORE JOINT BEHAVIOUR OF COUPLES OF VARIABLES (X,Y)

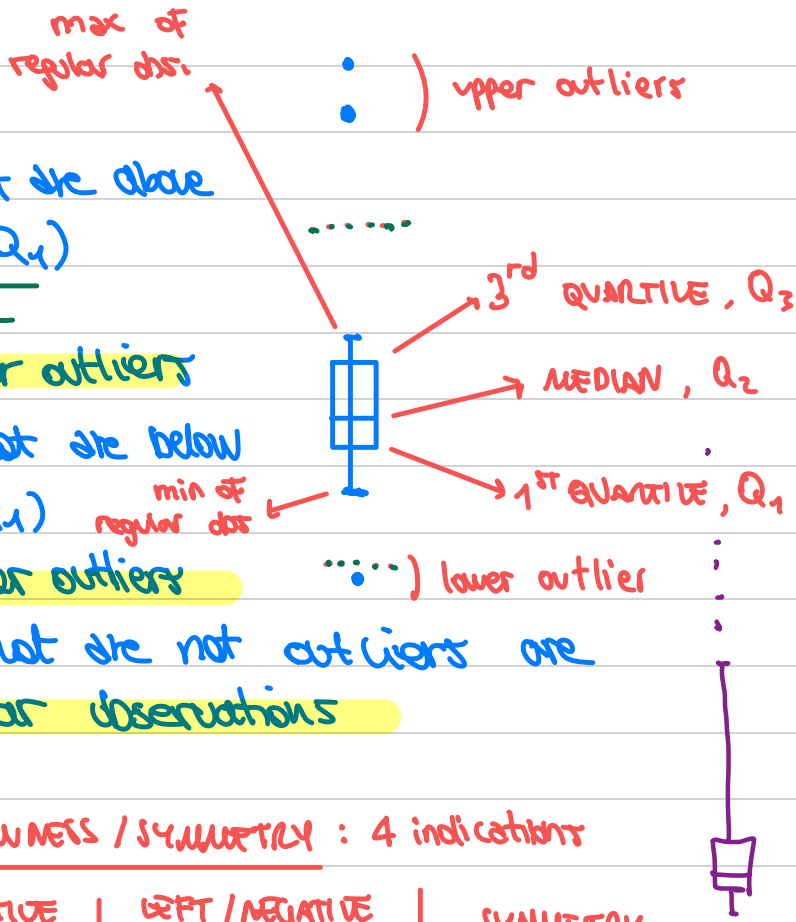
X \ Y	CATEGORICAL	NUMERICAL
CATEGORICAL	• cross table • stacked barplot	• multiple boxplot
NUMERICAL	• multiple boxplot	• scatterplot • correlation coefficient

CORRELATION COEFFICIENT X,Y := $\frac{1}{n} \frac{\sum_{i=1}^{n} (x_i - \text{MEAN}(x))(y_i - \text{MEAN}(y))}{SD(x)\, SD(y)}$

- measure the strength and direction of linear association (i.e. how closely points in the corresponding scatterplot follow a straight line )
- $-1 \leq r \leq 1$ always
- the higher $|r|$, the stronger the linear association
- high correlation does not imply causation
- low correlation does not imply independence

VISUALISATION OF SINGLE NUMERICAL VARIABLE

BOXPLOT



- Observations that are above $Q_3 + 1.5\,\underbrace{(Q_3 - Q_1)}_{IQR}$ are called upper outliers
- Observations that are below $Q_1 - 1.5\,(Q_3 - Q_1)$ are called lower outliers
- Observations that are not outliers are called regular observations

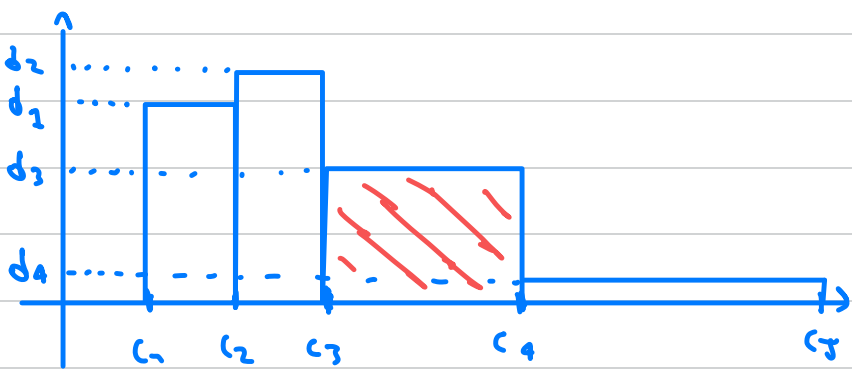
DISTRIBUTION SKEWNESS / SYMMETRY : 4 indications

	RIGHT / POSITIVE SKEW	LEFT / NEGATIVE SKEW	SYMMETRY
1	$(Q_3 - Me) > (Me - Q_1)$	$(Q_3 - Me) < (Me - Q_1)$	$(Q_3 - Me) \approx (Me - Q_1)$
2	upper whisker > lower whisker	upper whisker < lower whisker	upper whisker ≈ lower whisker
3	more upper outliers wrt lower outliers	more lower outliers wrt upper outliers	similar number of lower and upper outliers
4	Mean > Median	Mean < Median	Mean ≈ Median

VISUALISATION OF SINGLE NUMERICAL VARIABLE

HISTOGRAM

classes	abs. freq.	$p_k = \frac{f_k}{n}$ rel. freq.	$w_k = c_{k+1} - c_k$ class widths	$d_k = \frac{p_k}{w_k}$ freq. densities	$m_k = \frac{c_k + c_{k+1}}{2}$ mid. points
$[c_1, c_2)$	f_1	p_1	w_1	d_1	m_1
$[c_2, c_3)$	f_2	p_2	w_2	d_2	m_2
$[c_3, c_4)$	f_3	p_3	w_3	d_3	m_3
$[c_4, c_5)$	f_4	p_4	w_4	d_4	m_4



- The heights of a bar is the frequency density of the corresponding class. This way the area of a bar corresponds to the class relative frequency
- When the classes have same width, we can also use absolute or relative freq. as bar heights.

- From a histogram, or a distribution table for data grouped in classes, we can also get an approximation of the sample summary statistics such as mean, quantiles, variance and standard deviation.
- For example from the previous table

$$\text{MEAN} \approx \frac{1}{n} \sum_k f_k m_k$$

$$\text{VARIANCE} \approx \frac{1}{n} \sum_k (m_k - \text{MEAN})^2 f_k$$

MEDIAN $\approx$ value that splits the histogram in two region of same area

⋮

- All the APPROXIMATIONS above are obtained by assuming that all data are UNIFORMELY DISTRIBUTED within each interval class